

LAPORAN PRAKTIKUM PEMROGRAMAN WEB

Modul 10: Javascript Lanjutan



Oleh:

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A. TUJUAN PRAKTIKUM

Setelah menyelesaikan modul ini, peserta diharapkan mampu:

1. Memahami konsep dasar JavaScript sebagai bahasa pemrograman web.
2. Memahami fitur Modern JavaScript (ES6+) dan perannya sebagai fondasi React.
3. Mampu menggunakan sintaks ES6+ secara tepat dan konsisten.
4. Memahami konsep state, immutability, dan functional programming.
5. Mampu membangun aplikasi sederhana berbasis state menggunakan Vanilla JavaScript.

B. ALAT DAN BAHAN

Untuk menjalankan modul ini, diperlukan komputer atau laptop dengan sistem operasi Windows, Linux, atau macOS, minimal prosesor 1 GHz, RAM 2GB (disarankan 4GB atau lebih), serta browser modern seperti Google Chrome, Mozilla Firefox, atau Microsoft Edge.

Perangkat lunak yang digunakan meliputi:

- Code editor (Visual Studio Code, Sublime Text, atau sejenisnya)
- Web browser

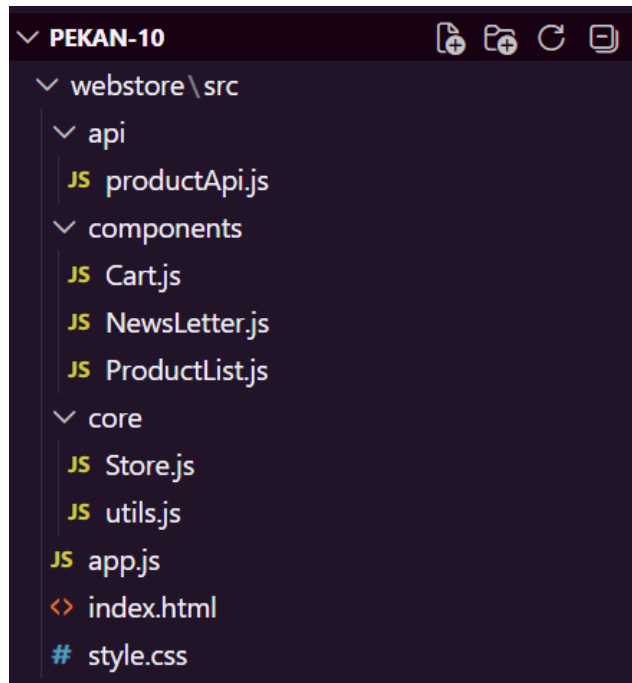
Bahan yang dibutuhkan:

- File HTML dasar
- File JavaScript (.js)

C. LANGKAH LANGKAH PRAKTIKUM

1) Studi Kasus 1

- a) Membuat folder dengan format atau penempatan seperti ini:



Gambar C.1.1 Format Folder

```
webstore > src > api > JS productApi.js > ...
1  export const fetchProducts = async () => {
2      return new Promise((resolve) => {
3          setTimeout(() => {
4              resolve([
5                  {
6                      id: 1,
7                      title: "Mechanical Keyboard",
8                      price: 150,
9                      rating: { rate: 4.8, count: 230 },
10                     image: "https://images.unsplash.com/photo-1618384887929-16ec33fab9ef?auto=format&fit=crop&w=800&q
11                 },
12                 {
13                     id: 2,
14                     title: "Gaming Mouse",
15                     price: 80,
16                     rating: { rate: 4.6, count: 150 },
17                     image: "https://images.unsplash.com/photo-1527814050087-3793815479db?auto=format&fit=crop&w=800&q
18                 },
19                 {
20                     id: 3,
21                     title: "4K Monitor 24\"",
22                     price: 300,
23                     rating: { rate: 4.9, count: 85 },
24                     image: "https://images.unsplash.com/photo-1527443224154-c4a3942d3acf?auto=format&fit=crop&w=800&q
25                 },
26                 {
27                     id: 4,
28                     title: "Premium Coffee Beans",
29                     price: 25,
```

```

webstore > src > api > JS productApi.js > ...
1  export const fetchProducts = async () => {
2      return new Promise((resolve) => {
3          setTimeout(() => {
20              {
27                  id: 4,
28                  title: "Premium Coffee Beans",
29                  price: 25,
30                  rating: { rate: 4.7, count: 500 },
31                  image: "https://images.unsplash.com/photo-1559056199-641a0ac8b55e?auto=format&fit=crop&w=800&q=80",
32              },
33              {
34                  id: 5,
35                  title: "Travel Tumbler",
36                  price: 35,
37                  rating: { rate: 4.5, count: 120 },
38                  image: "https://images.unsplash.com/photo-1578500494198-246f612d3b3d?auto=format&fit=crop&w=800&q=80",
39              },
40              {
41                  id: 6,
42                  title: "Noise Cancelling Headphones",
43                  price: 200,
44                  rating: { rate: 4.8, count: 320 },
45                  image: "https://images.unsplash.com/photo-1505740420928-5e560c06d30e?auto=format&fit=crop&w=800&q=80",
46              },
47          ]);
48      }, 800);
49  });
50  };

```

Gambar C.1.2 Code productApi.js

```

webstore > src > components > JS Cart.js > Cart > cart.map() callback
1  export function Cart(cart, onCheckout) {
2      const container = document.getElementById("cart");
3      container.innerHTML = "<h2>Your Cart</h2>";
4      if (cart.length === 0) {
5          container.innerHTML += "<p class='empty-msg'>Cart is empty.</p>";
6          return;
7      }
8      const list = document.createElement("div");
9      list.className = "cart-list";
10     cart.map((item) => {
11         const div = document.createElement("div");
12         div.className = "cart-item";
13         div.innerHTML = `
14             
15             <div class="cart-details">
16                 <h4>${item.title}</h4>
17                 <span>${item.price}</span>
18             </div> `;
19         list.appendChild(div);
20     });
21     container.appendChild(list);
22     const total = cart.reduce((acc, item) => acc + (item.price || 0), 0);
23     const footer = document.createElement("div");
24     footer.className = "cart-footer-summary";
25     footer.innerHTML = `
26         <div class="total-row">
27             <span>Total:</span>
28             <span class="total-price">${total.toFixed(2)}</span>
29         </div>

```

```

29     </div>
30     <button class="btn-primary" id="checkout-btn">Buy Now</button> `;
31     footer.querySelector("#checkout-btn").addEventListener("click", onCheckout);
32     container.appendChild(footer);
33 }
34

```

Gambar C.1.3 Code Cart.js

```

webstore > src > components > JS Newsletter.js > Newsletter > $content
1  export function Newsletter() {
2      const $container = $('<div class="newsletter-section"></div>');
3      const $content = $('`
4          <div class="newsletter-content">
5              <h3>Subscribe to our updates</h3>
6              <p>Get the latest tech news and offers.</p>
7              <form id="newsletter-form">
8                  <input type="email" placeholder="Enter your email" required />
9                  <button type="submit" class="btn-primary">Subscribe</button>
10             </form>
11             <div class="success-message" style="display:none; margin-top: 10px; color: green;">
12                 Thanks for subscribing!
13             </div>
14         </div>`;
15     );
16     $container.append($content);
17     $("#app").append($container);
18     $content.find("form").on("submit", function (e) {
19         E.preventDefault();
20         const $form = $(this);
21         $form.slideUp(400, function () {
22             $content.find(".success-message").fadeIn();
23         });
24     });
25 }

```

Gambar C.1.4 Newsletter.js

```

webstore > src > components > JS ProductList.js > ProductList > products.map() callback
1  rt function ProductList(products, onAdd) {
2    const container = document.getElementById("product-list");
3    container.innerHTML = "";
4    products.map((product) => {
5      const card = document.createElement("div");
6      card.className = "product-card";
7      const discount = Math.floor(Math.random() * 20) + 5;
8      const rating = product.rating || { rate: 4.5, count: 100 };
9      card.innerHTML = `
10     <div class="card-image">
11       
12     </div>
13
14     <div class="card-content">
15       <span class="badge">Up to ${discount}% off</span>
16       <a href="#" class="card-title" title="${product.title}">${product.title}</a>
17       <div class="rating-row">
18         <div class="stars">
19           ${getStarSVG().repeat(5)}
20         </div>
21         <span class="rating-score">${rating.rate}</span>
22         <span class="rating-count">(${rating.count})</span>
23       </div>
24       <ul class="features-list">
25         <li class="feature">
26           <svg xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">
27             <path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round"
28               strokeWidth="2" d="M13 7h6l2 4m-8-4v8m0-8V6a1 1 0 0 1-1-1H4a1 1 0 0 1-1-1V9h2m8 0H9m4 0h2m4 0h2v-4m0 0h-5m3.5

```

```

webstore > src > components > JS ProductList.js > getStarSVG
1  export function ProductList(products, onAdd) {
4    products.map((product) => {
31      Fast Delivery
32    </li>
33    </ul>
34    <div class="card-footer">
35      <span class="price">${product.price}</span>
36      <button class="add-btn btn-primary">
37        <svg xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">
38          <path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round"
39            strokeWidth="2" d="M4 4h1.5L8 16m0 0h8m-8 0a2 2 0 1 0 0 4 2 2 0 0 0 0-4Zm8 0a2
40            2 0 1 0 0 4 2 2 0 0 0 0-4Zm.75-3H7.5M11 7H6.312M17 4v6m-3-3h6" />
41          </svg>
42          Add
43        </button>
44      </div>
45    </div> `;
46    card.querySelector(".add-btn").addEventListener("click", function () {
47      onAdd(product);
48    });
49    container.appendChild(card);
50  });
51 }
52 function getStarSVG() {
53   return `<svg xmlns="http://www.w3.org/2000/svg" fill="currentColor" viewBox="0 0 20 20">
54     <path d="M9.049 2.927c-.921 1.603-.921 1.902 0 1.902 3.292a1 1 0 0 0 0.95.69h3.462c.969 0 1.371 1.24.588 1.811-2.8 2.
55     1.688-1.54 1.1181-2.8-2.034a1 1 0 0 0-1.175 0l-2.8 2.034c-.784.57-1.838-.197-1.5391.11811.07-3.292a1 1 0 0 0-.364-1.1
56   </svg>`;
57 }

```

Gambar C.1.5 ProductList.js

```

webstore > src > core > JS Store.js > ...
1  import { deepClone } from "../utils.js";
2
3  let state = {
4    products: [],
5    cart: [],
6  };
7
8  export const getState = () => deepClone(state);
9
10 export const setState = (newState) => {
11   const nextState = { ...state, ...newState };
12
13   state = nextState;
14
15   console.log("[Store] State updated:", state);
16 };

```

Gambar C.1.6 Store.js

```

webstore > src > core > JS utils.js > deepClone
1  import { getState } from "../Store.js";
2  export function deepClone(obj) {
3    if (obj === null || typeof obj !== "object") {
4      return obj;
5    }
6    if (obj instanceof Date) {
7      return new Date(obj.getTime());
8    }
9    if (Array.isArray(obj)) {
10     return obj.map((item) => deepClone(item));
11   }
12   const clonedObj = {};
13   for (const key in obj) {
14     if (Object.prototype.hasOwnProperty.call(obj, key)) {
15       clonedObj[key] = deepClone(obj[key]);
16     }
17   }
18   return clonedObj;
19 }

```

Gambar C.1.7 utils.js

```
webstore > src > JS app.js > checkout
1  import { fetchProducts } from "./api/productApi.js";
2  import { getState, setState } from "./core/Store.js";
3  import { ProductList } from "./components/ProductList.js";
4  import { Cart } from "./components/Cart.js";
5  import { Newsletter } from "./components/Newsletter.js";
6  function addToCart(product) {
7      const state = getState();
8      setState({ cart: [...state.cart, product] });
9      render();
10 }
11 function checkout() {
12     setState({ cart: [] });
13     alert("Thank you for your purchase!");
14     render();
15 }
16 function render() {
17     const state = getState();
18     ProductList(state.products, addToCart);
19     Cart(state.cart, checkout);
20 }
21 async function init() {
22     try {
23         const data = await fetchProducts();
24         setState({ products: data });
25         render();
26         Newsletter();
27     } catch (error) {
28         console.error("Failed to load products:", error);
29     }
}
```

Gambar C.1.8 app.js


```
webstore > src > index.html > html > body > div#app > div.container
```

```

1 <!doctype html>
2 <html lang="id">
3   <head>
4     <meta charset="UTF-8" />
5     <title>Vanilla JS Web Store</title>
6     <Link rel="preconnect" href="https://fonts.googleapis.com" />
7     <Link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
8     <Link
9       href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600&family=Merriwe+ather:ital,wght@0,400;0,70
10      rel="stylesheet"
11    />
12    <Link rel="stylesheet" href="style.css" />
13    <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
14    <script type="module" src="app.js" defer></script>
15  </head>
16  <body>
17    <div id="app">
18      <div class="container">
19        <div class="listing-header">
20          <nav class="breadcrumb" aria-label="Breadcrumb">
21            <a href="#">
22              <svg
23                xmlns="http://www.w3.org/2000/svg"
24                fill="currentColor"
25                viewBox="0 0 20 20"
26              >
27                <path
28                  d="m19.707 9.293-2-2-7-7a1 1 0 0 0-1.414 0l-7 7-2a1 1 0 0 0 1 1.414L2 10.414V18a2 2 0 0 0 2 2
29                />

```

```
webstore > src > index.html > html > body > div#app > div.container
```

```

2      <html lang="id">
16      <body>
17      <div id="app">
18      <div class="container">
19      <div class="listing-header">
20      <div class="listing-item">
21      <div class="listing-item">
22      <div class="listing-item">
23      <div class="listing-item">
24      <div class="listing-item">
25      <div class="listing-item">
26      <div class="listing-item">
27      <div class="listing-item">
28      <div class="listing-item">
29      <div class="listing-item">
30      </svg>
31      Home
32      </a>
33      <svg
34      style="color: #9ca3af"
35      xmlns="http://www.w3.org/2000/svg"
36      fill="none"
37      viewBox="0 0 24 24"
38      >
39      <path
40      stroke="currentColor"
41      stroke-linecap="round"
42      stroke-linejoin="round"
43      stroke-width="2"
44      d="m9 5 7 7 7 7"
45      />
46      </svg>
47      <a href="#">Products</a>
48      <svg
49      style="color: #9ca3af"
50      xmlns="http://www.w3.org/2000/svg"
51      fill="none"
52      viewBox="0 0 24 24"

```

```

webstore > src > index.html > html > body > div#app > div.container
 2  <html lang="id">
16  <body>
17    <div id="app">
18      <div class="container">
19        <div class="listing-header">
54          <path
55            stroke="currentColor"
56            stroke-linecap="round"
57            stroke-linejoin="round"
58            stroke-width="2"
59            d="m9 5 7 7 7 7"
60          />
61        </svg>
62        <span style="color: var(--text-muted)">Electronics</span>
63      </nav>
64      <div class="toolbar">
65        <h2 class="page-title">Electronics</h2>
66        <div class="toolbar-actions">
67          <button class="btn-secondary">
68            <svg
69              xmlns="http://www.w3.org/2000/svg"
70              fill="none"
71              viewBox="0 0 24 24"
72            >
73              <path
74                stroke="currentColor"
75                stroke-linecap="round"
76                stroke-width="2"
77                d="M18.796 4H5.204a1 1 0 0 0-.753 1.659l5.302 6.058a1 1 0 0 1 .247.659v4.874a.5.5 0 0 0 .2413 2
78

```

```

webstore > src > index.html > html > body > div#app > div.container
 2  <html lang="id">
16  <body>
17    <div id="app">
18      <div class="container">
19        <div class="listing-header">
78          />
79        </svg>
80        Filters
81      </button>
82      <button class="btn-secondary">
83        <svg
84          xmlns="http://www.w3.org/2000/svg"
85          fill="none"
86          viewBox="0 0 24 24"
87        >
88          <path
89            stroke="currentColor"
90            stroke-linecap="round"
91            stroke-linejoin="round"
92            stroke-width="2"
93            d="M7 4v16M7 4l3 3M7 4 7m9-3h6l-6 6h6m-6.5 10 3.5-7 3.5 7M14 18h4"
94          />
95        </svg>
96        Sort
97      </button>
98    </div>
99  </div>
100 </div>
101 <div id="product-list" class="product-grid"></div>

```

```

102     <div id="cart" class="cart-section"></div>
103   </div>
104 </div>
105   <script type="module" src="app.js"></script>
106 </body>
107 </html>
108

```

Gambar C.1.9 index.html

```

webstore > src > # style.css > #app
1  :root {
2
3    --primary: #0D9488;
4    --primary-hover: #0F766E;
5    --primary-light: #CCFBF1;
6    --bg-color: #F9FAFB;
7    --card-bg: #FFFFFF;
8    --text-main: #111827;
9    --text-muted: #6B7280;
10   --border-color: #E5E7EB;
11   --rating-star: #FACC15;
12   --font-serif: 'Merriweather', serif;
13   --font-sans: 'Inter', sans-serif;
14   --radius: 8px;
15 }
16
17 * {
18   box-sizing: border-box;
19   margin: 0;
20   padding: 0;
21 }
22
23 body {
24   font-family: var(--font-sans);
25   background: var(--bg-color);
26   color: var(--text-main);
27   line-height: 1.5;
28   -webkit-font-smoothing: antialiased;
29 }
30
webstore > src > # style.css > #app
31 .container {
32   max-width: 1280px;
33   margin: 0 auto;
34   padding: 32px 16px;
35 }
36
37 .flex-row,
38 .breadcrumb,
39 .breadcrumb a,
40 .toolbar,
41 .toolbar-actions,
42 .feature,
43 .rating-row,
44 .card-footer {
45   display: flex;
46   align-items: center;
47 }
48
49 .btn-primary,
50 .btn-secondary {
51   display: inline-flex;
52   align-items: center;
53   justify-content: center;
54   gap: 8px;
55   border-radius: var(--radius);
56   font-size: 0.875rem;
57   font-weight: 500;
58   cursor: pointer;
59   transition: 0.2s;
60 }

```

webstore > src > # style.css > #app

```
62  svg {
63    width: 16px;
64    height: 16px;
65  }
66
67  .listing-header {
68    margin-bottom: 32px;
69  }
70
71  .page-title {
72    font-family: var(--font-serif);
73    font-size: 1.875rem;
74    font-weight: 700;
75  }
76
77  .toolbar {
78    justify-content: space-between;
79    flex-wrap: wrap;
80    margin-top: 16px;
81    gap: 16px;
82  }
83
84  .toolbar-actions {
85    gap: 12px;
86  }
87
88  .breadcrumb {
89    gap: 8px;
90    font-size: 0.875rem;
```

webstore > src > # style.css > #app

```
88  .breadcrumb {
91    color: var(--text-muted);
92    margin-bottom: 12px;
93  }
94
95  .breadcrumb a {
96    color: var(--text-muted);
97    text-decoration: none;
98    gap: 6px;
99  }
100
101  .breadcrumb a:hover {
102    color: var(--primary);
103  }
104
105  .btn-secondary {
106    background: white;
107    border: 1px solid var(--border-color);
108    color: var(--text-main);
109    padding: 8px 12px;
110  }
111
112  .btn-secondary:hover {
113    background: #F3F4F6;
114    color: var(--primary);
115  }
116
117  .btn-primary {
118    background: var(--primary);
```

webstore > src > # style.css > #app

```
117  .btn-primary {
118    color: white;
119    border: none;
120    padding: 10px 20px;
121  }
122
123  .btn-primary:hover {
124    background: var(--primary-hover);
125  }
126
127  .btn-primary svg {
128    width: 20px;
129    height: 20px;
130  }
131
132
133  .product-grid {
134    display: grid;
135    grid-template-columns: repeat(auto-fill, minmax(250px, 1fr));
136    gap: 24px;
137    margin-bottom: 32px;
138  }
139
140  .product-card {
141    background: var(--card-bg);
142    border: 1px solid var(--border-color);
143    border-radius: var(--radius);
144    overflow: hidden;
145    display: flex;
146    flex-direction: column;
147    transition: box-shadow 0.2s
```

webstore > src > # style.css > #app

```
140  .product-card {
148    box-shadow: 0 1px 2px 0 rgba(0, 0, 0, 0.05);
149  }
150
151  .product-card:hover {
152    box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
153  }
154
155  .card-image {
156    height: 220px;
157    padding: 24px;
158    display: flex;
159    justify-content: center;
160    background:
161      white;
162  }
163
164  .card-image img {
165    max-height: 100%;
166    max-width: 100%;
167    object-fit: contain;
168  }
169
170  .card-content {
171    padding: 24px;
172    display: flex;
173    flex-direction: column;
174    flex: 1;
175  }
```

```

webstore > src > # style.css > % #app
177 .badge {
178     background: var(--primary-light);
179     color: var(--primary);
180     font-size: 0.75rem;
181     font-weight: 600;
182     padding: 2px 10px;
183     border-radius: 4px;
184     width: fit-content;
185     margin-bottom: 12px;
186 }
187
188
189 .card-title {
190     font-size: 1.125rem;
191     font-weight: 600;
192     color: var(--text-main);
193     text-decoration:
194         none;
195     margin-bottom: 8px;
196     line-height: 1.4;
197     display: -webkit-box;
198     -webkit-line-clamp: 2;
199     line-clamp: 2;
200     -webkit-box-orient: vertical;
201     overflow: hidden;
202 }
203
204
205 .card-title:hover {
206     text-decoration: underline;
207 }
208
209 .rating-row {
210     gap: 8px;
211     margin-bottom: 16px;
212 }
213
214 .stars {
215     display: flex;
216     color: var(--rating-star);
217 }
218
219 .rating-score {
220     font-size: 0.875rem;
221     font-weight: 600;
222 }
223
224 .rating-count,
225 .feature {
226     font-size: 0.875rem;
227     color: var(--text-muted);
228 }
229
230 .features-list {
231     display: flex;
232     gap: 12px;
233     margin-bottom: 20px;
234 }

```

```

webstore > src > # style.css > % #app
236 .feature {
237     gap: 6px;
238 }
239
240 .card-footer {
241     margin-top: auto;
242     justify-content: space-between;
243 }
244
245 .price {
246     font-size: 1.5rem;
247     font-weight: 800;
248 }
249
250 #cart-sidebar {
251     background: white;
252     border-left: 1px solid var(--border-color);
253     padding: 24px;
254     width: 350px;
255     position: fixed;
256     right: 0;
257     top: 0;
258     bottom: 0;
259     box-shadow: -4px 0 16px rgba(0, 0, 0, 0.05);
260     overflow-y: auto;
261     transform: translateX(100%);
262     transition: transform 0.3s;
263     z-index: 50;
264     display: flex;
265     flex-direction: column;
266
267     #cart-sidebar.open {
268         transform: translateX(0);
269     }
270
271     #cart-sidebar h2 {
272         margin-bottom: 24px;
273         font-family: var(--font-serif);
274     }
275
276     .empty-msg {
277         color: var(--text-muted);
278         font-style: italic;
279     }
280
281     .cart-list {
282         flex: 1;
283         overflow-y: auto;
284         display: flex;
285         flex-direction: column;
286         gap: 16px;
287         margin-bottom: 24px;
288     }
289
290     .cart-item {
291         display: flex;
292         align-items: center;
293
294     }

```

```
webstore > src > # style.css > % #app
```

```
291 .cart-item {  
294   gap: 12px;  
295   padding-bottom: 12px;  
296   border-bottom: 1px solid var(--border-color);  
297 }  
298  
299 .cart-item img {  
300   width: 50px;  
301   height: 50px;  
302   object-fit: cover;  
303   border-radius: 4px;  
304   border: 1px solid var(--border-color);  
305 }  
306  
307 .cart-details h4 {  
308   font-size: 0.875rem;  
309   font-weight: 600;  
310   margin-bottom: 4px;  
311   display: -webkit-box;  
312   -webkit-line-clamp: 1;  
313   line-clamp: 1;  
314   -webkit-box-orient: vertical;  
315   overflow: hidden;  
316 }  
317  
318 .cart-details span {  
319   font-size: 0.875rem;  
320   color: var(--text-muted);  
321 }
```

```
webstore > src > # style.css > % #app
```

```
323 .cart-footer-summary {  
324   border-top: 2px solid var(--border-color);  
325   padding-top: 20px;  
326   margin-top: auto;  
327 }  
328  
329 .total-row {  
330   display: flex;  
331   justify-content: space-between;  
332   align-items: center;  
333   margin-bottom:  
334     16px;  
335   font-weight: 700;  
336   font-size: 1.125rem;  
337 }  
338  
339 .total-price {  
340   color: var(--primary);  
341 }  
342  
343 #checkout-btn {  
344   width: 100%;  
345 }  
346  
347 .newsletter-section {  
348   margin-top: 64px;  
349   background: var(--card-bg);  
350   border-top: 1px solid var(--border-color);  
351   padding: 48px 0;
```

```
webstore > src > # style.css > % #app
```

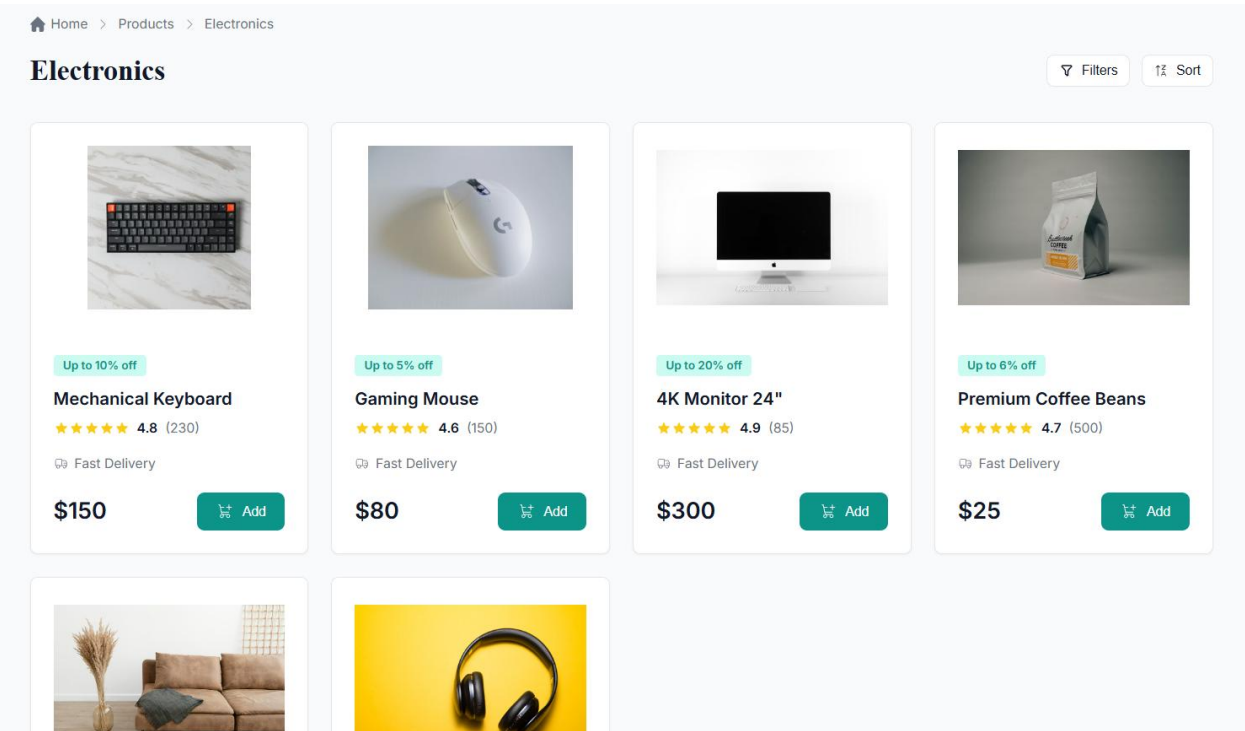
```
347 .newsletter-section {  
352   text-align: center;  
353 }  
354  
355 .newsletter-content {  
356   max-width: 500px;  
357   margin: 0 auto;  
358 }  
359  
360 .newsletter-content h3 {  
361   font-family: var(--font-serif);  
362   font-size: 1.5rem;  
363   margin-bottom: 8px;  
364 }  
365  
366 .newsletter-content p {  
367   color: var(--text-muted);  
368   margin-bottom: 24px;  
369 }  
370  
371 .newsletter-content form {  
372   display: flex;  
373   gap: 8px;  
374 }  
375  
376 .newsletter-content input {  
377   flex: 1;  
378   padding: 10px;  
379   border: 1px solid var(--border-color);
```

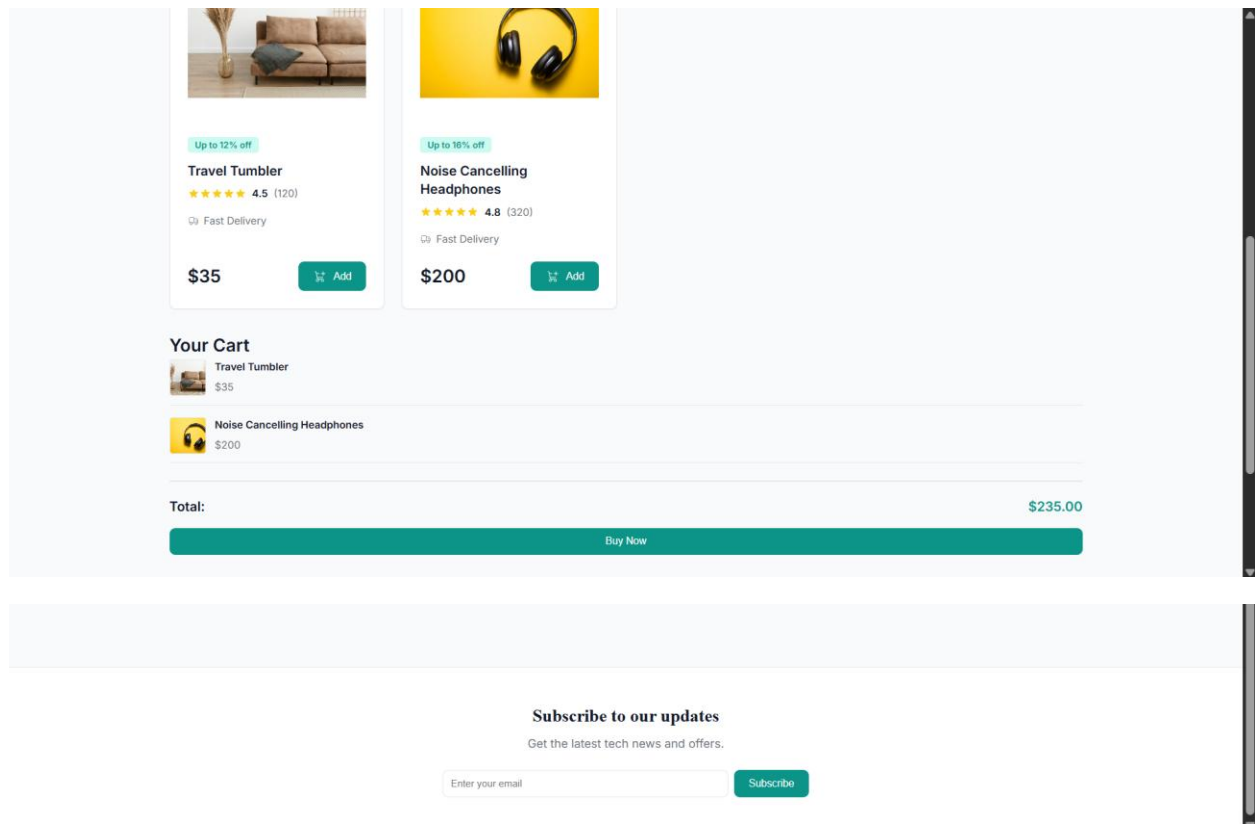
```
379     border: 1px solid var(--border-color);
380     border-radius: var(--radius);
381 }
382
383 #app {
384     width: 100%;
385 }
```

Gambar C.1.10 style.css

D. HASIL DAN DOKUMENTASI PROGRAM

1) Hasil Studi Kasus 1





E. ANALISIS DAN PEMBAHASAN

Tugas praktikum membangun sebuah webstore sederhana berbasis Vanilla JS yang mengintegrasikan seluruh konsep ES6+ yang telah dipelajari. Aplikasi ini dirancang dengan arsitektur berlapis yang terdiri dari folder `src/api` untuk pengambilan data, `src/core` untuk manajemen *state* dan utilitas, serta `src/components` untuk komponen UI. File `productApi.js` mensimulasikan pengambilan data produk dari server menggunakan Promise dan `setTimeout` yang bersifat asinkronus, sehingga mahasiswa dapat memahami bagaimana data nyata dari API bekerja sebelum ditampilkan ke halaman. Pendekatan modular ini memisahkan tanggung jawab setiap bagian kode secara jelas sehingga lebih mudah dirawat dan dikembangkan.

Manajemen *state* aplikasi dipusatkan di `Store.js` menggunakan dua fungsi utama yaitu `getState()` dan `setState()`. Yang menarik, fungsi `getState()` tidak langsung mengembalikan objek *state* asli, melainkan menggunakan fungsi `deepClone()` dari `utils.js` untuk mengembalikan salinan mendalam dari *state*. Hal ini memastikan bahwa komponen yang menerima *state* tidak dapat mengubah data asli secara langsung, sehingga prinsip *immutability* benar-benar diterapkan. Setiap kali tombol *Add* ditekan, fungsi `addToCart()` di `app.js` memanggil `setState()` dengan menyebarkan *state* lama menggunakan spread operator dan menambahkan produk baru, kemudian memanggil `render()` untuk memperbarui tampilan secara keseluruhan.

Pada sisi komponen, ProductList.js menggunakan `map()` untuk mengubah setiap data produk menjadi elemen kartu HTML secara dinamis, sementara Cart.js menggunakan `reduce()` untuk menghitung total harga seluruh item di keranjang. Komponen Newsletter.js memanfaatkan jQuery untuk animasi `slideUp` pada form dan `fadeIn` pada pesan sukses ketika pengguna berhasil berlangganan, memperlihatkan bagaimana jQuery dan JavaScript murni dapat digunakan bersama dalam satu proyek. Keseluruhan alur mulai dari pengambilan data, penyimpanan *state*, hingga render ulang komponen mencerminkan pola yang sangat mirip dengan cara kerja React, sehingga tugas ini berfungsi sebagai jembatan konseptual yang kuat sebelum mempelajari framework tersebut.

F. KESIMPULAN

Tugas praktikum webstore ini membuktikan bahwa konsep-konsep ES6+ seperti *async/await*, *immutability*, spread operator, `map()`, `reduce()`, *module system*, dan *closure* bukan hanya teori yang berdiri sendiri, melainkan saling bekerja sama membentuk sebuah aplikasi yang fungsional dan terstruktur. Arsitektur yang memisahkan data, logika *state*, dan tampilan UI mencerminkan pola pengembangan aplikasi modern yang profesional, sekaligus membekali mahasiswa dengan pemahaman konseptual yang kuat sebagai persiapan sebelum mempelajari framework React secara lebih mendalam.

G. LAMPIRAN

<https://github.com/AriqAhmadNayaka/WebPro4904.git>